Amirhosein Farahmand

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Education

B.Sc. in Computer Science 2021 - 2025

- University of Tehran. *Tehran, Iran*
- Last Two Years GPA: **18.6**/20 (WES: **3.85**/4).
- GPA: **17.74**/20 (WES: **3.62**/4).

Experience

Research Assistant — Clinical Prediction Modelling and Survival Analysis Mar. 2025 – Present
University of Tehran *Tehran, Iran*

- Developing and evaluating methods for estimating **time-dependent net benefit** for survival risk prediction models under **informative censoring**, covering both model-development and validation settings.
- Extended the framework to time-dependent and competing-risk outcomes and contributed to the `dcurves` R package for robust net-benefit estimation under informative censoring.
- Research abstract accepted by the **Society for Medical Decision Making** and selected for an **oral presentation** at its 2026 Annual Meeting.

Research Assistant — Multimodal AI and Large Language Models Jun. 2024 – May 2025
GISMA University of Applied Sciences *Potsdam, Germany (Remote)*

- Extracted textual and visual information from PDF reports to construct structured datasets from unstructured documents.
- Designed a **multimodal Retrieval-Augmented Generation (RAG)** system using **large language models** for claim verification across text and image inputs.
- Developed pipelines for evidence retrieval, multimodal analysis, and claim validation against a domain-specific knowledge base.
- Co-authored *Fact-Checking Sustainability Objectives Using Multimodal Retrieval-Augmented Generation*, accepted at the **IEEE International Conference on Data Mining 2025**.

Teaching Assistant — Computer Science, Statistics, and Machine Learning Sep. 2023 – Present
University of Tehran *Tehran, Iran*

- Designed assignments, course projects, and instructional sessions for undergraduate and graduate courses, including **Machine Learning**, **Stochastic Processes**, **Nonlinear Optimization**, **Multivariate Probability**, **Probability**, **Statistical Methods**, **Linear Algebra**, **Basic Programming**, and **Calculus II**.
- Delivered instructional sessions to more than **250 students** and prepared over **25 assignment sets and course projects**.
- Graded more than **1,000 assignments, quizzes, projects, and examinations**.

Publications

Robust Net Benefit Calculation for Survival Risk Prediction Models in the Presence of Informative Censoring 2025–Present

- Abstract accepted by the **Society for Medical Decision Making** and selected for an **oral presentation** at the 2026 Annual Meeting in Oslo. *(Abstract, Code)*
- Authors: **Amirhosein Farahmand**, Mohsen Sadatsafavi, Abdollah Safari, and Tae Yoon Lee.

Fact-Checking Sustainability Objectives Using Multimodal Retrieval-Augmented Generation 2025

- Accepted at the **IEEE International Conference on Data Mining 2025**. *(Paper, Code)*
- Authors: Mohammad Mahdavi, **Amirhosein Farahmand**, and Abolfazl Nadi.

A Q-Learning Approach to Maintenance Policy Optimization for Series Systems with Degradation and Common Shocks 2025

- Accepted at the **11th Seminar on Reliability Theory and Its Applications**, University of Isfahan. *(Proceeding, Code)*
- Authors: **Amirhosein Farahmand**, Fatemeh Fartash, Mohammadreza Yadollahi, and Sina Zarandi.

Selected Projects

Deep Generative Models for Image Modelling *(GitHub)* 2025

- Implemented and evaluated a range of deep generative models for image data, including **NADE**, **VAE**, **β-VAE**, and **Conditional VAE**.
- Developed **Generative Adversarial Networks**, **Energy-Based Models**, and **NICE normalizing flows**, and compared their generative performance.
- Implemented **DDPM** and **DDIM** diffusion models and explored distillation methods for accelerating image-generation sampling.
- Evaluated generated images using **Inception Score**, **Kernel Inception Distance**, and **Fréchet Inception Distance**.

Gaussian Mixture Model for Stress-Level Classification *(GitHub)* Jan. 2025

- Developed a Gaussian Mixture Model in a Bayesian framework to classify stress levels using physiological and contextual data from wearable sensors.
- Implemented the Expectation-Maximization algorithm for parameter estimation and evaluated model performance on the Nurse Stress Prediction dataset.

Relevant Coursework

- Genereative Models (A+)
- Non-linear Optimization (A+, Top Score)
- Artificial Intelligence (A+)
- Stochastic Processes (A+, Top Score)
- Linear Algebra (A+, Top Score)
- Scientific Computing (A+, Top Score)

Research Interests

- Machine Learning
- Survival Analysis
- Causal Inference
- AI for Healthcare

Skills

- **Programming Languages**: Python, R, C/C++
- **Frameworks/Libraries**: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, caret, ggplot2, tidyr, dplyr
- **Tools**: Git, GitHub/GitLab, Linux
- **Soft Skills**: Communication, Teamwork, Teaching
- **Languages**:
  - **English** — IELTS Overall Band 7
  - **Farsi** — Native